Effectiveness of the Serious Game 'Rejuvenesce Village' in Cognitive Rehabilitation for the Elderly

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ABSTRACT

The existing computer-assisted cognitive rehabilitation tools are almost training-purposed ones which the users may feel boring and are repetitive with no fun. The authors have developed a serious game called 'Rejuvenesce Village' for cognitive rehabilitation with fun and good feelings for the elderly. This study is for verifying the effectiveness of the serious game 'Rejuvenesce Village' in cognitive rehabilitation for the elderly. The experiments for the verification were performed for 5 months, in Seoul Asan Medical Center, numbering to 100 elderly participants. The results showed the improvement of the cognitive functions and also the improvement of life satisfaction and the decrease of depression.

Keywords: *Attention, Cognitive Function, Dementia, Depression, Elderly Game, IRB, Judgment, Memory, Reaction Rate, Serious Game*

1. INTRODUCTION

In accordance with prolonging the average life span of Korean people, the aging of the population structure was shown prominently. While it was the positive change due to the advances in medicine and the improvement of nutritional status, changes in the structure of the modern family and the composition changes of the population associated with low birth rates were causing the elderly problems in many respects. The elderly might experience the depression including loss of physical and cognitive function, due to the sociological alienation and biological aging (Y. Kang, 2005).

In this study, a serious game called 'Rejuvenesce Village' has been developed to improve the cognitive function and the life satisfaction of the elderly as well as to reduce depression through the training of the attention, memorization and judgment. Serious games claim to be more than just entertainment and pursue the

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serious aim to transfer knowledge in a playful way (Clark, 1970; F. Hauser, A. Leich & K. Schiffer, 2013)

In other words, it could be developed and applied according to the characteristics of the user individuals by integrating the benefits of various treatments and therapies and complementing the disadvantage.

Previous studies related to the game have been studied by causing the interest to induce voluntary and active participation and to maintain the motive in raising the confidence and a variety of effects such as mental comfort, intellectual function, and fun (J. Kwon at el, 2010; S. Kim, 2008; J. Kim, 2005; Y. Park & H. Kim, 2010; S. Seo, 2003). However, the studies on the application status of serious game and on the effect were still insufficient for the purpose of improving cognitive function, and there was insufficient data for content development (S. Shin, 2003). Thus, multi-disciplinary research was needed to fit the ultimate research purposes of the serious game improving cognitive function. The systematic analysis focused on the serious game to improve the cognitive function is needed especially among the effectiveness of validation studies for the elderly.

For this, the implementation of this study 'Rejuvenesce Village' as a tool for clinical trials were conducted from July 2012 for 5 months, in Asan Medical Center, in the target of 100 people aged over 65. The analysis of the experimental content had positive results in terms of life satisfaction and depression, and to develop effective elderly serious game, the game design requirements could be analyzed. As a result of study, the games and equipment were spread in the 25 dementia centers in Seoul in 2013.

2. GAME DEVELOPMENT AND EXPERIMENTS

2.1. Cognitive Impairment and Depression in the Elderly

The elderly were easy to cause accidents due to the slowing of the reaction rate such as exercise response, reaction movement, memory, information processing, etc. Unlike the changes due to normal aging, the problem of the elderly on other disorders such as vascular disease, trauma of cognitive function gave the patient as well as the elderly family member or guardian a physical, mental and financial burden (Y. Won & S. Park, 2011). Cognitive dysfunction entailed the severe loss of memory and problem-solving skills enough to cause significant changes compared to the functioning level in the past. Dementia was particularly a representative disease of cognitive dysfunction, on the rise in the elderly population, emerging the world's biggest health problem in the 21st century. Moreover, after 65 years old, the dementia was known to increase doubles every five years and a common disease in old age, even with the increase of the elderly population, the incidence of dementia was expected to increase continuously (Y. Lee, 2011).

The improvement of cognitive function and the decrease of depression had relationship. The most of dementia are in the irreversible nature and due to the main symptom such as depression and other behavioral disorders that is caused by cognitive impairment, the active supports for the patients as well as their families are necessary (M. Cho, 2009). The elderly depression

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are appearing more with increasing age, lower income, lower education levels, alone or in poor physical health, the poorer functional health such as activities of daily living. Therefore, the way to improve cognitive function and to reduce depression have been actively sought (S. Jo & H. Yoon, 2011).

Depending upon what do you do during leisure time in old age, the satisfaction of life will vary for more enjoyable and healthy life, the application of serious games for the elderly that using the computer program as a media have been tried (Y. Lee, J. Ahn & K. Lim, 2009).

2.2 The Project Requirements of the Serious Game for the Elderly

Aging, dementia, and stroke could cause declines or disabilities of cognitive functions as memory or attention (S. Bae, W. Jeong, H. Lee & K. Kim, 2012).

Computer-assisted cognitive rehabilitation of persons with brain injury could be effective to improve the cognitive functions as memory or attention (Chen S H et al., 1997; Moniz-Cook E & Wood R, 1997)

Base on the frame of reference for visual-perception (Miller J, 1982) and the principle plasticity of brain, the proposed computer-assisted cognitive rehabilitation was designed for improving the primary cognitive functions of memory and attention. And it's also for improving the higher-level ability of judgment and reaction with fun and good feelings.

The proposed computer-assisted cognitive rehabilitation consisted of the visual feedback, visual search, attention, information processing speed, memory, and problem-solving skills, and the flexibility and control of the treatment as well as the reduction of treatment time, objective measures of performance, and the immediate feedback to patients could be given (Y. Lee, J. Ahn & K. Lim, 2009).

The goal of our study is to find the source technologies of elderly serious game and to validate their actual effectiveness by measuring the reaction rates of dementia-related cognitive elements, memory, judgment, attention, and other medical data shows on Table 1.

By measuring the relationship between the time and accuracy of the user behavior and the time required to solve a given task, the phase memory, working memory, spatiotemporal attention, agility, and executing function can be measured, and their intercorrelations can be analyzed (Kielhofner G, 2002). The results of previous studies[1)] can be combined with this project to revive the 1970s' nostalgia and memories of the elderly who were in their 30s or 40s at that time, and to make this game more interesting. The color, interface, and screen layout need to be developed considering the physical characteristics of the elderly ("Korean Dementia Association", 2006).

The objectives of the game for the elderly, which induces their interests and cognitive ability, are to improve their learning through the fun game; to accelerate their brain activity, with the attention, memory, judgment, and reaction that match the cognitive function factors; to enhance their cognitive ability for each element via repetitive learning from watching the game results in real time and retrying for better results; and to recover their confidence and improve their social adaptability by improving their cognitive function (M. Hong, 2007).

The main element of the game for repeated training is the record breaking, and the real-time processing of the movement distance/time measurements, reaction time, and accuracy objectively and systematically induces challenges and retries to get better results, and allows the elderly to perform the challenges easily, with great interest.

The training objectives must be presented clearly and in a manner in which they will not be recognized as mere learning objectives. The approach should be easily accessible for the elderly so that they can recognize the learning activity as a pleasant activity, and enjoy it (Y. Hong & Y. Kang, 2009).

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Table 1. Planning approach to improve the cognitive function

1. Reaction Rate

- According to a variety of screen changes, screen touch and ball touch
- Development orientation : simplify the color, shape and apply to the intuitive UI

2. Judgement

- Confirm if possible to judge accurately rather than speed and quickness
- Development orientation : apply the function to measure the logic and thinking skills

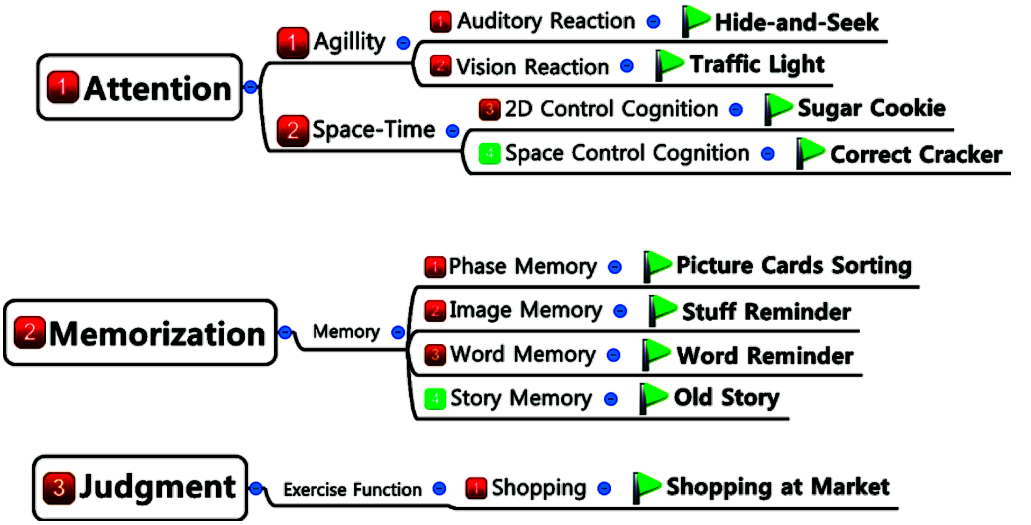
3. Attention

- Insert the inference function to things and situation
- Development Direction : directly reflect the blocked module and gamer's activities

4. Memory

- Insert the elements that gamers can remind and think
- Development orientation : the function to store the major info rmation through storage cell

Figure 1. Nine games of 'Rejuvenesce Village' categorized by the function



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Table 2. Game Overview to improve cognitive function for the Elderly

Purpose	Improving the cognitive function and life satisfaction, decrease of depression	
Control	Touch screen and Ball-button touch	
Target User	Main	Slight cognitive dysfunction elderly over 65 years old with a risk of dementia
	Sub	Mid, old generation needed training for cognitive function
Estimated Effect	Improvement of cognitive function and enhancement of coordination movement, the elderly game treatment-decrease of depression	
People	1 person	
Playtime	50-60 min for full function traning	

3. GAME DEVELOPMENT

3.1. Game Overview

In this paper, by applying components of the cognitive function for the purpose of rehabilitation, the elderly serious game 'Rejuvenesce Village' has been developed.

Key elements of cognitive function were attention, memorization, and judgment because aging, dementia, and stroke could usually cause declines or disabilities of memory, attention, or judgment for the elderly (S. Bae, W. Jeong, H. Lee & K. Kim, 2012; "Korean Dementia Association", 2006; M. Hong, 2007). The sub-item of the categories was implemented for the eight games and each game consisted of three levels of difficulty (Fig. 1).

The principle of the serious game was to compensate and replace the remaining normal cognitive function by the principle plasticity of brain through repeated training of damaged specific cognitive functions. Users through mini-games proceeded in combination to the key elements of basic training.

'Rejuvenesce Village' mind and body weakened more than 65 years of normal elderly with dementia severity of the risk, but is targeted at the elderly. To simplify the interface intuitive visual access of the elderly to enjoy the games difficult for the user interface of the appropriate size were considered. Hands, feet, etc. through the use of light body with improved cognitive function, exercise capacity improved coordination can be expected. (Table 2)

Elderly-friendly visual and color by applying traditional games were based on the configuration of the content. Bound to one of the stories in the form of a number of episodes of each game were to be linked in the 1970s, by configuring the rural town nostalgia of the elderly, and leveraging a thing of the past for them was constructed world view.

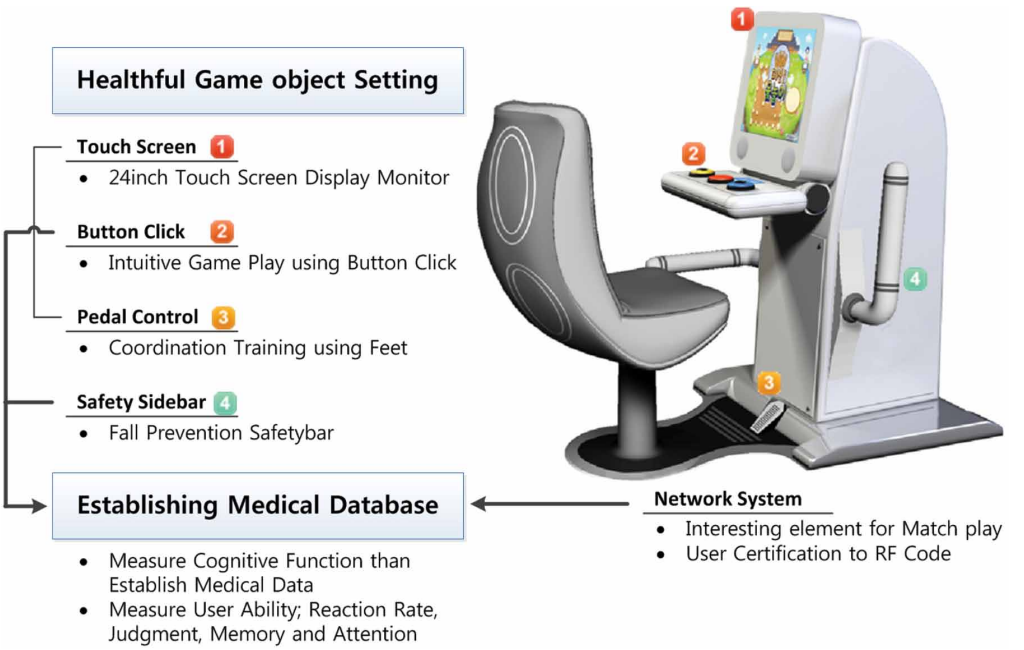
3.2. Game Interface

Elderly targeted to make the features of the game is appropriate for the nature of the interface has been developed. Seniors who are not familiar with state-of-the-art equipment used in the unit not inconvenient to implement, easy-to-learn about the operation is possible. Direct and easy access to the old man in action was implemented to increase, the ability to respond to accidents in most cases, so that the loss of the safety consideration was designed primarily. (Fig 2)

A touch screen and a button eye-hand coordination (hand-Eye Co-ordination) touch interface that can be trained directly on the

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Figure 2. Game interface



screen or press a hand to press the button with the physical activity of visual and auditory stimulation signal is a function of the training, each element can be naturally.

3.3. Visual requirements

3.3.1. GUI (Graphic User Interface) Size

In a GUI, the visibility and attractiveness of the texts and buttons must be emphasized. The normal text size for a screen 1 m away from a viewer is desirably 40-45 pt, and its actual size is 1.4x1.5-1.7x1.8 cm. The maximum number of characters in one text screen is desirably 14. The minimum height of the button text font is desirably 1.95-2.1 cm. The average distal finger segment is 2.0-2.5 cm long, and the minimum height of the button is desirably 4.0-4.5 cm assuming the finger touches the center of the button. The proper font size for buttons is 50-55 pt.

The square object size whose one side is 160-164 px is suitable for the elderly, and it can be adjusted by 5-10% considering the aged eye.

3.3.2 Color configuration

Notice the color of things, gender -based set of objects over the area to be smaller than the actual size may seem higher than the color saturation and brightness is appropriate to the target. Attention of the object if it needs to be emphasized sex chromatic than achromatic two, cold color disapproval than two, more saturated low saturation is effective. After hearing for inducing color configuration is applied to the object. Cold color colors calm and focused state of the system is induced, warm colors of the system leads to a state of tension and excitement. Gameplay of the player's excitement rises when using cold color system to induce sedation, the player will feel bored if the disapproval of the interval using the system to induce suspense and excitement is effective. In close proximity to each other between an object or objects and

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Figure 3. Clinical Trial at Seoul Asan Medical Center



the background by the saturation difference is greater than mutation can cause an optical illusion, because it is appropriate to add an outline. Game play time increases the longer the time, the proportion of cold color to feel a sense of focus to keep short to induce effective and fast-moving objects if the low saturation and low intensity, slow motion high saturation and high brightness to use are suitable.

4. EXPERIMENTS AND ANALYSIS

4.1. Overview of Experiments

In this paper, the proposed game design, how you feel about the actual interest and to evaluate the effectiveness of July 2012, 5 months, Asan

Table 3. Overview of experiments

Date	1. Jul. 2012 - 30. Nov. 2012
Place	Seoul Asan Medical Center, Seoul, Korea.
Objective	Improvement of cognitive function and life satisfaction, remission depression
Persons	100 elderly people over 65 age
Method	Examination and questionnaire of the interview when begin(end) of the experiment
Time	60 min per time, 2 times per week, total 8 times

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Table 4. The impact on the cognitive function

	Range	Before game - after game		t	p
		Average	Std. Deviation		
Cognitive Function	Man	-0.48	1.62	-1.61	.119
	Woman	-0.51	1.47	-2.90	.005
	Total	-0.5	1.51	-3.32	.001

Medical Center, 100 elderly subjects in clinical trials were conducted. (Fig 3, Table3)

4.2. Experimental Analysis

The elderly who participated in the test were 65-80 years old, 70.1 (±3.77) years old on average. They were classified into three categories: those who were 65-70 years old, accounting for 43% of the test participants; those who were 70-74 years old (42%); and those who were 75 years old or older (15%). The "young old" (those who were 65-74 years old) accounted for 85% of the test participants, and the others were "middle old." Fifty two percent of the participants exercised five times or more in a week, and 7% answered that they had never exercised. Forty five percent of the participants exercised less than 1-1.5 hours at a time.

Functional categories of pre-game cognitive function score was 24-30 points, on average 27.6 (± 1.70), the mean function score after the game is 28.1 (± 1.75) was statistically no significant difference in cognitive function after a game high emerged (t = -3.32, p = .001). (Table 4)

Functional categories of pre-game depression score was 0-15 points, on average 3.3 (± 3.07) points, respectively. Depression scores after the game that the category of 0-14 and the mean3.0 (± 3.03) point's game ever appeared depressed after the game the score was low. However, when comparing before and after the

Table 5. The Effect on the depression

	Range	Before game - after game		t	p
		Average	Std. Deviation		
Depression	Man	0.34	2.39	0.78	.445
	Woman	0.20	2.23	0.75	.458
	Total	0.24	2.27	1.06	.292

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Table 6. The impact on the life satisfaction

	Range	Before game - after game		t	p
		Average	Std. Deviation		
Life Satisfaction	Man	-0.48	1.27	-2.05	.050
	Woman	-0.18	1.33	-1.16	.252
	Total	-0.27	1.32	-2.05	.043

game showed no significant difference statistically (t = 1.06, p = .292). (Table 5)

Game of life satisfaction score of all functional categories were 3-11 points, on average 6.5 (± 1.62) points, respectively. After the game the category of life satisfaction score was 3-11 points, on average 6.8 (± 1.81) points after the game as the game of life satisfaction scores ever was higher. When comparing before and after the game there were significant differences statistically (t = -2.05, p = .043). (Table 6)

Satisfaction with game content as a font size of 99% of the subjects responded that the current was moderate. Voice coming out of the game early, 70% of subjects in the current opinion is reasonable, but wishing little faster and 22%, respectively.60% of the subjects in the speed game, the game difficulty is 65% of the subjects responded that this was appropriate.

5. CONCLUSION

In this paper, we proposed the effective serious game in cognitive rehabilitation for the elderly in order to give the users fun and good feelings. We show that the serious game is an effective cognitive rehabilitation tool by the experiment results. Also it is effective for improving life satisfaction and decreasing the depression.

The clinical experiments targeted to 100 seniors was performed. As results of the experiments, the satisfaction of the game showed satisfaction levels as a whole; it could obtain the conclusion that was appropriate. However, the life satisfaction and depression items showed significant difference, we could know that the game could not affect the whole life of the elderly.

Future research is needed to analyze the effects of cognitive function by item and long-term clinical trials. If the subdivision applied to serious game was analyzed, the new study results will be able to be obtained.

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